

3.5 Transmit Squelch Architecture (Pre-Driver, Driver, MRM, MRM Controller)

3.5.1 Function and normative basis

Transmit squelch is the modulation-suppressed, power-preserving TX state mandated by TXO-010 (OCI Table 2-2, note 4): launched OMA ≤ -15 dBm per channel while average launched power remains constant. It is not merely a quiet mode - it is a protocol signal. The OCI deskew state machine (Section 1.1) uses squelch as the relink handshake: entering Deskew_Data_Relink squelches modulation on all four channels of a 212.5 Gb/s lane for relink_squelch_tx_duration (60-75 ms, Table 1-3) to command the far end to restart its deskew process, after which the transmitter proceeds directly into the 160-bit deskew training pattern. The constant-average-power constraint exists because both ends of the link hold micro-ring resonances (TX modulator rings and RX demux filter rings) locked using average-power-derived error signals; a squelch that disturbed average power would unlock heaters at both ends and turn a 75 ms handshake into a full re-acquisition. Squelch is therefore a whole-signal-chain behavior spanning the pre-driver, driver, MRM, and MRM controller, and it is deliberately distinct from PMD transmit disable (DJI-003), which removes optical power entirely via the ELS path for safety and management purposes.

3.5.2 Squelch principle for a micro-ring transmitter

In mission mode the MRM converts driver swing into optical amplitude by moving the ring resonance between two detuning states, producing levels P1 and P0; the launched average power is approximately $(P1+P0)/2$ for balanced data. Squelch must freeze the output at that average while removing the swing. Two levers accomplish this: (1) the electrical modulation is muted in the pre-driver/driver (swing suppression), and (2) the MRM is held at a calibrated static bias V_{sq} chosen so that the ring's static transmission equals the time-averaged transmission of the modulated state. Because the ring transfer function is strongly nonlinear in both voltage and wavelength, V_{sq} is not the electrical mid-swing point and cannot be derived analytically with sufficient accuracy: it is a per-channel calibrated quantity, trimmed against the TX monitor photodiode (SQL-006). Two second-order effects must be absorbed by the MRM controller: removing RF modulation changes the average intracavity optical energy and the carrier-induced self-heating of the junction, producing a small resonance step at squelch entry and exit (SQL-007/009). The heater servo, whose error signal is the monitor-photodiode average power - which squelch by definition preserves - remains closed-loop throughout and absorbs this step.

3.5.3 Block-level responsibilities

Pre-driver.

- Data-path source select: a glitch-free three-way multiplexer (mission data / deskew training-release pattern / squelch hold code), switched synchronously to the serializer clock. The TX PLL and serializer never stop; only the data content changes, guaranteeing the phase-continuity the deskew protocol requires of pattern transitions (SQL-003).
- Mute control: primary swing suppression is commanded here (swing-DAC to minimum or data forced to the hold code), sequenced with the driver mute to reach the composite ≥ 20 dB electrical suppression target (SQL-001).

- Common-mode and balance hold: the pre-driver output common mode and duty-cycle balance are held through mute entry/exit so no common-mode step propagates to the modulator and perturbs AOP beyond the ± 0.5 dB window (SQL-004).

Driver.

- Independent swing and bias paths: the output stage shall separate the RF swing (tail current / swing DAC) from the DC bias network applied across the MRM junction, so that muting the swing leaves the modulator DC operating point untouched (SQL-005). A topology in which swing and bias share a control (e.g., supply-collapse mute) is prohibited because it moves the ring operating point and violates the constant-AOP requirement.
- Mute depth and settling: driver mute contributes the balance of the ≥ 20 dB electrical suppression, with entry/exit settling ≤ 1 ms so the driver transient is negligible against the 60 ms minimum squelch dwell (SQL-005).
- Residual-swing floor: driver output-stage feedthrough (clock, data-path leakage) at the launched-OMA level must remain below -15 dBm optical; this sets a driver isolation/feedthrough specification derived in the DTX-012 allocation.

Micro-ring modulator (MRM).

- Calibrated squelch bias V_{sq} per channel, applied through the driver DC path, placing static transmission at the mission-mode average (SQL-006). Calibration is factory-set and periodically re-trimmed in service using the monitor photodiode, since V_{sq} drifts with ring detuning, temperature, and aging.
- Operating-regime containment: V_{sq} shall keep the junction in its normal (depletion or injection) regime so carrier-plasma index/absorption changes between squelched and modulated states stay small and the resulting resonance step remains inside the heater servo's linear capture range (SQL-007).

MRM controller (heater servo).

- Continuous closed-loop lock: the servo's error signal is average power at the monitor photodiode, which squelch preserves by construction - this is precisely why TXO-010 forbids average-power squelch. The loop, including its lock dither, runs uninterrupted through entry, dwell, and exit (SQL-008).
- Dither budget in squelch: heater dither converts to residual OMA at the output; dither amplitude during squelch shall be bounded so dither-induced OMA does not consume the -15 dBm squelched-OMA limit (SQL-008). If mission-mode dither is too large, a reduced-dither squelch profile shall be defined - with the corresponding lock-slope penalty analyzed.
- Thermal-step absorption: the servo shall settle the squelch entry/exit resonance step to within the mission detuning budget in ≤ 10 ms (target), comfortably inside the 60 ms minimum dwell, with zero unlock events over the qualification soak (SQL-009).
- Far-end dependency: the same constant-AOP property keeps the remote RX demux filter servo locked (DRX-008); any AOP droop during our squelch is a direct attack on the partner's lock margin, which is why SQL-002 is budgeted at ± 0.5 dB rather than at the 3 dB LOS margin.

3.5.4 Entry/exit sequence

- Entry (t_0): freeze heater-servo integrator ramp limits (transient guard), switch pre-driver source to hold code, ramp driver swing to mute, apply V_{sq} trim - completed within 1 ms; AOP excursion within ± 0.5 dB.

- Dwell (t0 to t0+60..75 ms): servo re-settles thermal step (≤ 10 ms), dither continues at squelch profile; residual OMA verified ≤ -15 dBm; far end detects loss of modulation within its 50 ms t_loselock budget and enters relink.
- Exit (t0+relink_squelch_tx_duration): pre-driver source switches phase-continuously to the deskew training pattern, driver swing ramps to mission amplitude, V_sq trim released to mission bias - CDR-safe, glitch-free, and immediately compliant with the ≥ 285 ms training-pattern transmission requirement (SQL-010).
- Abort/fault: if the servo reports unlock during dwell, the channel escalates to TX fault (REG-007) rather than exiting squelch with an unlocked ring.

3.5.5 Derived requirements

Req ID	Requirement	Source	Ver.
SQL-001	In squelch mode, launched OMA per channel shall be ≤ -15 dBm while modulation is suppressed; the pre-driver/driver combination shall provide ≥ 20 dB of electrical swing suppression (design target), yielding ≥ 14 dB optical OMA suppression relative to the maximum operating OMA of -1 dBm with margin.	TXO-010; OCI T.2-2 n.4	T
SQL-002	Average launched optical power per channel during squelch shall remain constant within ± 0.5 dB (implementation target) of the pre-squelch mission-mode value, so that far-end demux-filter heater lock is maintained and far-end LOS (assert max -14 dBm AOP) is never triggered by a squelch event.	TXO-010; RXO-006	T
SQL-003	The pre-driver shall implement a glitch-free three-way data-path source select (mission data / deskew pattern / squelch hold) with the serializer and TX PLL running continuously in all modes; source transitions shall be phase-continuous so the far-end CDR does not lose lock.	LOG-003; DTX-012	T/D
SQL-004	The pre-driver mute path shall hold its output common-mode voltage and duty-cycle balance during mute entry/exit; any common-mode step at the driver input shall settle such that the launched AOP transient stays within the SQL-002 window.	DTX-012	T
SQL-005	The driver shall implement swing mute (tail-current or swing-DAC reduction) while independently preserving the DC bias applied across the MRM junction; the modulator DC operating point shall not move when the RF swing is removed. Mute entry/exit settling shall be ≤ 1 ms (target).	DTX-012; TXO-010	T
SQL-006	A per-channel calibrated squelch bias point V_sq shall be applied to the MRM such that the static optical transmission equals the time-averaged transmission of the modulated state (the ring transfer function is nonlinear: V_sq is not the electrical mid-swing). V_sq shall be factory-calibrated and periodically re-trimmed against the monitor photodiode.	TXO-010; DTX-012	T/A
SQL-007	The MRM junction shall remain in its normal depletion/injection operating regime at V_sq (no forward-bias excursion, no regime change), bounding the carrier-induced index and absorption	DTX-004/005	A/T

Req ID	Requirement	Source	Ver.
	difference between squelched and modulated states so the residual resonance shift is within the heater servo's linear capture range.		
SQL-008	The MRM controller (heater servo) shall remain in closed-loop lock throughout squelch entry, dwell, and exit, using the monitor-photodiode average-power error signal; lock dither shall continue during squelch with amplitude bounded such that dither-induced OMA does not violate SQL-001.	TXO-010; DTX-005	T
SQL-009	The heater servo shall absorb the thermal step caused by removal/restoration of RF modulation (change in average intracavity energy and carrier heating) without an unlock event, settling residual detuning to within the mission-mode detuning budget in ≤ 10 ms (target), i.e., well inside the 60 ms minimum relink squelch window.	DTX-004/005; LOG-004	T/A
SQL-010	Squelch entry/exit sequencing shall satisfy OCI Table 1-3: squelch applied for relink_squelch_tx_duration (60-75 ms) before deskew training-pattern transmission begins; exit shall proceed directly and glitch-free into the 160-bit training pattern; the far end shall be able to detect loss of modulation within its 50 ms t_{loselock} budget, which bounds allowable residual OMA ripple during dwell.	LOG-004; BUP-004	T
SQL-011	Squelch shall be architecturally distinct from PMD transmit disable: transmit disable (management/safety path, DJI-003) removes optical output via ELS attenuation/shutdown and does NOT preserve average power; squelch preserves average power and shall never be implemented by ELS power reduction, heater park, or laser blanking.	DJI-003; REG-002; TXO-010	I/T
SQL-012	All four channels of a 212.5 Gb/s lane shall enter and exit squelch together (lane-atomic), with channel-to-channel entry/exit alignment such that the far-end deskew engine observes a consistent lane state; per-channel squelch is reserved for test modes only.	OCI Sec.1.1	T

3.5.6 Calibration and verification hooks

- Per-channel V_{sq} calibration routine (factory and in-service re-trim) using the TX monitor photodiode; stored per channel with temperature annotation.
- Squelched-OMA compliance measurement at TP2 (≤ -15 dBm) and AOP-delta measurement across entry/dwell/exit (± 0.5 dB window), per channel, across temperature corners.
- Heater-lock soak: 10k+ squelch entry/exit cycles at hot and cold plate corners with zero unlock events and logged residual-detuning settling profiles (feeds VER-006/VER-008).
- Partner-survival test: far-end CDR and demux-filter lock monitored through squelch cycles (loss-of-modulation detect ≤ 50 ms, no LOS assertion, no far-end heater unlock) - the system-level proof of SQL-002/003 (feeds VER-008).

- Disable-vs-squelch discrimination test: verify PMD transmit disable and squelch produce their distinct optical signatures (AOP removed vs. AOP preserved) and cannot be cross-triggered (SQL-011).